CASE STUDY 4 – EMPLOYEE PAYROLL AND DEDUCTION ANALYSIS

1. Problem Statement

Calculate gross salary from basic pay, allowances, overtime, and incentives; apply progressive deductions and generate a formatted salary statement.

2. Objectives

- Calculate allowances, overtime, incentives, and gross salary.
- Apply deductions according to salary bands.
- Calculate net salary.
- Print a readable payroll statement for each employee.

3. Assumptions

- HRA is 20% of basic pay; travel allowance is 8% of basic pay.
- Overtime pay is ₹200 per hour; incentive is supplied per employee.
- Deduction is 5% of gross salary up to ₹25,000, 10% on the portion from ₹25,001 to ₹50,000, and 15% on the portion above ₹50,000.
- All amounts are in Indian rupees.

4. Algorithm

- Store employee details as tuples.
- Calculate HRA, travel allowance, overtime, and gross salary.
- Calculate progressive deduction by salary bands.
- Subtract deduction from gross salary to get net salary.
- Display each salary statement.

5. Pseudocode

START

Store employee payroll records.

FOR each employee: calculate HRA, travel allowance, and overtime pay.

Calculate gross salary.

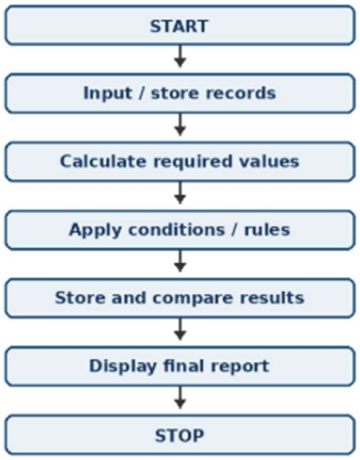
Calculate progressive deduction on each salary band.

Calculate net salary = gross salary - deduction.

Display salary statement.

STOP

6. Flowchart



7. Python Source Code

```
employees = [
    ("Asha", 22000, 5, 1500),
    ("Balu", 35000, 8, 2500),
    ("Charan", 50000, 10, 4000)
]
for employee in employees:
    name, basic, overtime_hours, incentive = employee
    hra = basic * 0.20
    travel = basic * 0.08
    overtime = overtime_hours * 200
    gross = basic + hra + travel + overtime + incentive
    if gross <= 25000:
        deduction = gross * 0.05
    elif gross <= 50000:
        deduction = 25000 * 0.05 + (gross - 25000) * 0.10
    else:
```

```
        deduction = 25000 * 0.05 + 25000 * 0.10 + (gross - 50000) * 0.15
    net = gross - deduction
    print("EMPLOYEE:", name)
    print("Basic:", round(basic, 2), "| HRA:", round(hra, 2), "| Travel:",
round(travel, 2))
    print("Overtime:", round(overtime, 2), "| Incentive:", round(incentive, 2))
    print("Gross:", round(gross, 2), "| Deduction:", round(deduction, 2), "| Net:",
round(net, 2))
    print("----------------------------------------")
```

8. Test Cases

Test Case	Input / Condition	Expected Result
1	Basic ₹22,000; 5 overtime hours; incentive ₹1,500	Gross ₹30,660; deduction ₹1,816; net ₹28,844
2	Basic ₹35,000; 8 overtime hours; incentive ₹2,500	Gross ₹48,900; deduction ₹3,640; net ₹45,260
3	Basic ₹50,000; 10 overtime hours; incentive ₹4,000	Gross ₹70,000; deduction ₹6,750; net ₹63,250

9. Sample Output Screenshot

```
EMPLOYEE: Asha
Basic: 22000 | HRA: 4400.0 | Travel: 1760.0
Overtime: 1000 | Incentive: 1500
Gross: 30660.0 | Deduction: 1816.0 | Net: 28844.0
----------------------------------------
EMPLOYEE: Balu
Basic: 35000 | HRA: 7000.0 | Travel: 2800.0
Overtime: 1600 | Incentive: 2500
Gross: 48900.0 | Deduction: 3640.0 | Net: 45260.0
----------------------------------------
EMPLOYEE: Charan
Basic: 50000 | HRA: 10000.0 | Travel: 4000.0
Overtime: 2000 | Incentive: 4000
Gross: 70000.0 | Deduction: 6750.0 | Net: 63250.0
----------------------------------------
```

10. Conclusion

The program generates payroll statements and calculates progressive deductions. It demonstrates arithmetic expressions, conditional bands, tuple records, unpacking, and iteration.